GF(27) in GALG and FFGFT:

Why G(3) Contains No GF(27) but G(6) Does

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Abstract

Doug Matzke reports in his IPI mail of 1 September 2026 that a systematic search in GALG G(3) (6561 attempts) finds no element of order 26. This document explains algebraically why, shows where such elements exist in G(6), and draws the consequences for the FFGFT mass layer.

Three theorems: **(A)** G$(3) \cong M_2$(GF$(9))^2$: order 26 is algebraically impossible in G(3) — not a search error, but a theorem. **(B)** G$(6) \cong M_8$(GF(9)): order-26 elements exist; a 5-blade example can be verified directly in GALG. **(C)** GF(27) is present in G(6) as an order structure, not as a subfield (since $3 \nmid 8$) — exactly the FFGFT statement from Doc. 338.

Consequence: $1/\alpha = 3700/27$ lives on the G(3)/GF(9) level; the lepton mass layer $(m_\mu/m_e)^2 = 43200$ needs G(6) with order-26 elements. Verification script: `pruef_333_gf27_in_galg.py` (all assertions **[B]**).

.1 Occasion and Starting Point

Doug Matzke (IPI, 1 Sept. 2026) finds in GALG G(3): GF(9) in the generator $(a + ja)^8 = 1$; GF(81) in G(3) with $(1 + a + b + c + (a \wedge b))^{80} = 1$; orders $\{2, 3, 4, 8, 16, 32, 40, 80\}$. An explicit search for order 26:

```
gasolve([a,b,c], is_root26): Attempted 6561 with 0 found.
```

This is not a search error. It is an algebraic theorem.

.2 Theorem A: G(3) Contains No GF(27)

Structure of G(3)

GALG G(n) = Cl(n) over the symmetric Z3C = GF(9) = GF$(3)[i]$, where $i^2 = -1$, characteristic 3. For odd $n = 3$:

$$G(3) = \mathrm{Cl}(3) \cong M_2(\mathrm{GF}(9)) \oplus M_2(\mathrm{GF}(9)) \tag{1}$$

Every irreducible representation is 2-dimensional over GF(9).

Eigenvalue Structure

The eigenvalues of a 2×2 matrix over GF(9) lie in the algebraic closure — in the smallest extension field GF(9^2) = GF(81).

$$|\mathrm{GF}(81)^*| = 80 = 2^4 \cdot 5 \tag{2}$$

The number 13 does not divide 80. Therefore:

- No element of M_2(GF(9)) has an order divisible by 13.
- Order 26 requires $13 \mid$ ord(X) — impossible in G(3).
- Doug's search space $\{a, b, c\}$ spans G(3) — the result 0/6561 is algebraically forced **[B]**.

Consistency with Doug's Findings

Table 1: Orders in GALG G(3) and their Galois origin

Order	Origin	Divides
2	$\lvert$GF$(3)^*\rvert$	80
4	$\varphi(5) = 4$	80
5	min. prime factor $\lvert$GF$(81)^*\rvert$	80
8	$\lvert$GF$(9)^*\rvert$	80
16, 20, 40, 80	subgroups of GF$(81)^*$	80
3	unipotent (char. 3)	—
26	$\lvert$GF$(27)^*\rvert$, needs factor 13	**not** 80

A numerical sample of 3000 random Cl(3) elements confirms: no order is divisible by 13 **[B]**.

.3 Theorem B: G(6) Contains Order-26 Elements

Structure of G(6)

$$G(6) = \mathrm{Cl}(6) \cong M_8(\mathrm{GF}(9)) \tag{3}$$

Irreducible representations are 8-dimensional over GF(9).

Construction via Irreducible Polynomial

The polynomial $f(x) = x^3 + 2x + 1$ over GF(3) (equivalent to $x^3 - x + 1$) has no root in GF(3) and is therefore irreducible. Since gcd$(3, 2) = 1$, f remains irreducible over GF(9); its roots lie in GF$(27) \setminus$ GF(3). The companion matrix

$$C_f = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathrm{GF}(3)) \tag{4}$$

has order 26 in GL$_3$(GF(3)) (f is primitive) **[B]**.

Embedding in G(6) as a GALG Element

The 8×8 element $X = C_f \oplus C_f \oplus (-I_2)$ satisfies:

- $X^{26} = 1$, $X^{13} \neq 1$, $X^2 \neq 1$ — order exactly 26 **[B]**
- Blade decomposition over 38 Z3C blades from $\{e_1, \dots, e_6\}$

Minimal Example: 5 Blades, Directly Verifiable in GALG

A systematic search shows: order-26 elements need at least 5 blades; with 4 blades there are none. The following 5-blade element can be verified directly in Doug's GALG tool:

$$X = -(1+i)(e_1e_2) + (1+i)(e_1e_2e_3e_6) + (1-i)(e_1e_2e_5e_6) + (1+i)(e_1e_4e_5e_6) - (e_1e_2e_3e_4e_6) \tag{5}$$

GALG syntax:

```
X = -(1+1j)*(e1^e2) + (1+1j)*(e1^e2^e3^e6)
+ (1-1j)*(e1^e2^e5^e6) + (1+1j)*(e1^e4^e5^e6)
- (e1^e2^e3^e4^e6)
X**26 == 1 # True
X**13 == 1 # False
```

[B] (pruef_333, all assertions passed)

Frequency in G(6)

Table 2: Fraction of elements with $13 \mid \mathrm{ord}(X)$ in G(6) by blade support

Support (number of blades)	Hits / Invertible	Fraction
≤ 4	0 / ~450	0 %
5	rare	< 1 %
6	85 / 478	17.8 %
10	190 / 511	37.2 %

Order-26 elements are common in G(6) — Doug's search in G(3) (span$\{a, b, c\}$) could not find them for two reasons: algebraically (Theorem A) and because G$(3) \subsetneq G(6)$.

.4 Theorem C: GF(27) as Order Structure, not Subfield

Subfield Criterion

A unital embedding GF$(27) \hookrightarrow M_n$(GF(9)) makes GF$(9)^n$ a GF$(27) \otimes$ GF(9) = GF(729)-vector space. Since [GF(729) : GF$(9)] = 3$, we need $3 \mid n$.

Table 3: GF(27) as subfield in M_n(GF(9))

n	$3 \mid n$	Subfield possible?
2	no	no (G(3))
8	no	no (G(6))
3	yes	yes
6	yes	yes
9	yes	yes

For G(6) with $n = 8$: no GF(27) subfield. But the minimal polynomial of X factors:

$$\chi_X(x) = (x^3 + 2x + 1)^2 \cdot (x + 1)^2 \quad \in \text{GF}(3)[x] \tag{6}$$

Two cubic factors — eigenvalues in GF(27) — and one linear factor from GF(3). GF(27) appears as splitting field, not as subalgebra.

Connection to Docs. 330/338

This is exactly the FFGFT statement from Docs. 330 and 338:

$$\text{GF}(9) \not\subset \text{GF}(27) \quad (\text{since } \gcd(2,3) = 1) \tag{7}$$

Both fields sit in parallel in GF(3^6) = GF(729), not in a tower. The mass-ratio identity $(m_\mu/m_e)^2 = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|$ uses both fields *in parallel* — consistent with Theorem C.

.5 Consequences for FFGFT and GALG

Table 4: GF(27) structure in GALG and FFGFT

Level	GALG	FFGFT	Status
α	G(3)/GF(9)	$1/\alpha = 3700/27$, cancels	**[B]**
Lepton masses	G(6), ord. 26 needed	$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$	**[K]**
GF(27) subfield	absent in G(3), G(6)	parallel in GF(729)	**[B]**

The fine-structure constant $1/\alpha = 3700/27$ lives on the GF(9)/G(3) level: $|\text{GF}(27)| = 27$ cancels in the derivation. Doug's statement "I'm not working on mass" and his search result 0/6561 are both consistent with this layering:

- G(3): GF(9), GF(81), α-structure — Doug's working level **[B]**
- G(6): order-26 elements, lepton mass layer — FFGFT level **[K]**

Verification Script

`pruef_333_gf27_in_galg.py` — all three theorems verified numerically ($Cl(6)$ over $GF(9)$ as 8×8 matrices; exact order determination; blade decomposition with Z3C coefficients; samples of 3000 and 600 elements). All assertions passed **[B]**.

Note on AI Assistance

This document was prepared with support from Claude (Anthropic). All algebraic computations were verified by Python scripts with exact arithmetic over $GF(9)$ as 8×8 matrix representations. The physical interpretations and conclusions are by Johann Pascher (ORCID 0009-0000-6518-4064).

Bibliography

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